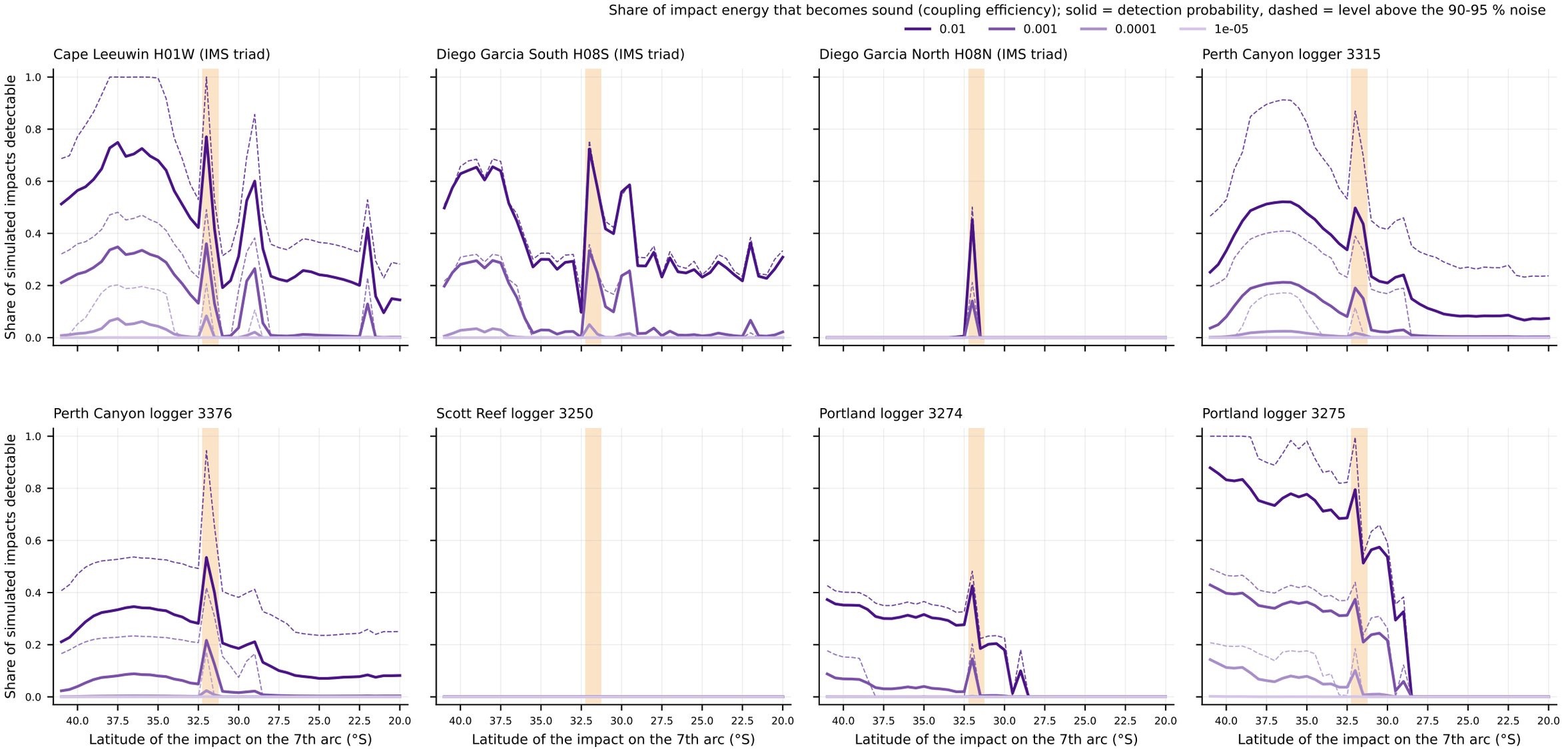
Share of simulated impacts that would be detectable at each receiver: appreciable only for a coupling efficiency of 0.001 or more, and smaller north of 30°S



Plain-language note: For an impact at each latitude on the 7th arc, the chart shows the share of simulated impacts whose sound would be strong enough to detect at the receiver, for four assumed values of the share of impact energy that becomes sound. The impact energies come from the project's end-of-flight simulation (run J). The dashed line uses a stricter rule: the sound must be above the loudest 5-10 % of the background noise. In the orange band the source sits over Broken Ridge (water less than 2,000 m deep), where the model is not validated. These are model predictions, not detections; the sound-loss model is not yet calibrated.

Technical note: Source posterior run J (one seed per stratum, equal stratum weights, prior weight = 00:19 held out, n = 20000 draws of E = contact kinetic energy and tau90; tau_ac = tau90 clamped 0.05-10 s, k = 1). Received 5-40 Hz peak 1 s SPL = sum_b eta E rho c/(2 pi) S_b(f_c, 1/tau) 10^(-TL_b/10) / T_eff, T_eff = 2.68 s (T_C2), f^-2 spectrum, TL at 10 m source from the grid (IMS: KRAKEN - RAM delta; IMOS: RAM-corrected). Solid: mean stage-A logistic P_D(SNR) (3376 fit for Perth Canyon, 3274 fit for Portland, 3250 fit for Scott Reef; IMS BORROWED 3274 fit + 4.8 dB triad gain), SNR against the noise median (IMOS: imos2b background median; IMS: 3376 proxy + Blackman App. B offset, digitised 2003). Dashed: share of draws above the noise p95 (IMOS, imos1d 1 s percentiles) or p90 (IMS, Blackman panel spread). eta = 1e-2 and 1e-3 exceed the F-35A upper anchor 2.1e-4 and are shown as the swept range asked for. PREDICTIVE, NOT CALIBRATED; caveats C1, C2. Grid: passes A+B (offsets -30/0/+30 NM, mandated) and C (±60 NM, supplementary); module commits fc825c0, 4392091.